

**UNIT-IV****LINEAR ORDINARY DIFFERENTIAL EQUATIONS OF HIGHER ORDER****Topic Learning Objectives:**

- Understand the formulation of higher-order linear ordinary differential equations with constant coefficients and the concept of their general solution.
- Develop the ability to solve homogeneous differential equations by determining complementary functions using characteristic equations.
- Gain knowledge of nonhomogeneous differential equations and methods to obtain particular integrals using inverse differential operators and appropriate techniques based on the given forcing function.
- Apply the method of variation of parameters to obtain particular solutions of nonhomogeneous differential equations.
- Understand and solve differential equations with variable (functional) coefficients, particularly Cauchy (Euler) equations.
- Interpret and model real-world physical problems such as simple harmonic motion and LRC circuits using higher-order differential equations.

Differential equations arise in an attempt to describe physical phenomena in mathematical terms. A single differential equation can serve as a mathematical model for many different processes. In the study of mechanics – spring/mass systems, the free, underdamped and critically damped motion can be expressed as linear differential equation. Similar to this, LRC series circuits can be described by a linear second- order differential equation.

**Differential Equation:**

An equation containing the derivatives of one or more dependent variables, with respect to one or more independent variables, is said to be differential equation.

The ‘order’ of a differential equation is the order of the highest derivative in the equation and the power of highest order derivative after clearing fractional powers from derivatives in the equation gives the ‘degree’ of differential equation.

**Classification of differential equation:**

Differential equations are classified as:

- i. Ordinary differential equation (ODE)
- ii. Partial differential equations (PDE)

If the differential equation contains ordinary derivatives of one or more dependent variables with respect to single independent variable, equation is known as ODE.

**Examples:**

- i)  $\ddot{x} = -n^2x$
- ii)  $\frac{[1+(y')^2]^{3/2}}{y''} = a$

If the differential equation contains partial derivatives of one or more dependent variables with respect to two or more independent variables, equation is known as PDE.

**Examples:**

i)  $u_{xx} + u_{yy} = 0$

ii)  $c^2 u_{xx} = u_t$

### **Linear Differential Equation:**

A differential equation is said to be linear if dependent variable, all its derivatives are of first degree and they are not multiplied together.

A non-linear differential equation is one that is not linear.

### **Linear ODE of higher order:**

A linear ODE of nth order is of the form:

$$(a_0 D^n + a_1 D^{n-1} + a_2 D^{n-2} + \dots + a_{n-1} D + a_n)y = g(x) \quad \dots (1),$$

where  $a_0, a_1, \dots, a_n$  are either constants or function of  $x$  alone,  $D = \frac{d}{dx}$  differential operator and  $g(x)$  is a known function of  $x$ .

Let  $F(D) = (a_0 D^n + a_1 D^{n-1} + a_2 D^{n-2} + \dots + a_{n-1} D + a_n)$  be a differential operator, then above equation can be written as:

$$F(D)y = g(x) \quad \dots (2)$$

If  $g(x) = 0$ , equation is said to be 'homogeneous' otherwise it is known as 'non-homogeneous'. If  $y_1, y_2, \dots, y_n$  are linearly independent solutions of the  $n^{\text{th}}$  order homogeneous differential equation then  $y = c_1 y_1 + c_2 y_2 + \dots + c_n y_n$ , where  $c_1, c_2, \dots, c_n$  are arbitrary constants is also a solution of homogeneous equation.

If  $y_p$  be a particular function which satisfies non-homogeneous linear differential equation then the general solution of the equation (2) is given by:

$$y = y_c + y_p$$

Where  $y_c = c_1 y_1 + c_2 y_2 + \dots + c_n y_n$  is called as complementary function and  $y_p$  is called particular integral.

### **Homogeneous Linear ODE with constant coefficients:**

Consider a nth order homogeneous ODE,

$$(a_0 D^n + a_1 D^{n-1} + a_2 D^{n-2} + \dots + a_{n-1} D + a_n)y = 0$$

Let  $y = e^{mx}$  be a trial solution, obtaining the derivatives and using above equation we obtain:

$$e^{mx}(a_0 m^n + a_1 m^{n-1} + a_2 m^{n-2} + \dots + a_{n-1} m + a_n) = 0$$

Since  $e^{mx} \neq 0$ , above equation vanishes if 'm' is root of algebraic equation. This algebraic equation is known as 'auxiliary' or 'characteristic' equation. The general solution of homogeneous equation depends on nature of roots of this equation.

Consider the second order ODE of the form:

$$(a_0 D^2 + a_1 D + a_n)y = 0$$

The auxiliary equation is given by  $a_0 m^2 + a_1 m + a_n = 0$ .

**Case (i):** Let the roots of auxiliary equation be real and distinct say

$m = m_1, m_2$ . The general solution of above equation is given by

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$$

**Case (ii):** Let the roots of auxiliary equation be real and repeated say  $m = m_1, m_1$ . The general solution of above equation is given by

$$y = (c_1 + c_2 x) e^{m_1 x}$$

**Case (iii):** Let the roots of auxiliary equation be complex say

$m = \alpha + i\beta, m = \alpha - i\beta$ . The general solution of above equation is given by  $y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$

**Case (iv):** If the ODE is fourth order and the roots of auxiliary equation are complex repeated say  $m = \alpha \pm i\beta, m = \alpha \pm i\beta$ . The general solution of equation is given by:

$$y = e^{\alpha x} [(c_1 + c_2 x) \cos \beta x + (c_3 + c_4 x) \sin \beta x]$$

The above procedure can be extended to higher order linear homogenous ODE in a similar manner.

Higher-order linear differential equations with constant coefficients model systems where the output depends on multiple rates of change, and their homogeneous solutions (complementary functions) describe the natural behaviour of systems without external input. In computer science, they are used in system modelling, control systems, and signal processing, where they help analyse stability, transient response, and behaviour of algorithms and dynamic systems such as filters, circuits, and feedback systems.

### Examples:

1. Solve:  $2 \frac{d^2 y}{dx^2} - 5 \frac{dy}{dx} - 3y = 0$

**Solution:** The auxiliary equation is given by  $2m^2 - 5m - 3 = 0$

The roots are  $m = -\frac{1}{2}, m = 3$ . The roots are real and distinct the general solution of given ODE is

$$y = c_1 e^{-\left(\frac{1}{2}\right)x} + c_2 e^{3x}$$

2. Solve:  $\frac{d^2 y}{dx^2} - 10 \frac{dy}{dx} + 25y = 0$

**Solution:** The auxiliary equation is given by  $(m - 5)^2 = 0$

The roots are  $m = 5, 5$  as the roots are real and repeated the general solution of given ODE

$$\text{is } y = c_1 e^{5x} + c_2 x e^{5x}$$

3. Solve:  $\frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} + 7y = 0$

**Solution:** The auxiliary equation is given by  $m^2 + 4m + 7 = 0$

The roots are  $m = -2 \pm \sqrt{3}i$  which are complex, hence the general solution is given by  $y = e^{-2x} [c_1 \cos(\sqrt{3}x) + c_2 \sin(\sqrt{3}x)]$

4. Solve:  $(D^3 + 1)y = 0$

**Solution:** The auxiliary equation is given by  $m^3 + 1 = 0$

The roots are  $m = -1, \frac{1}{2} \pm i \frac{\sqrt{3}}{2}$ , hence the general solution is given by

$$y = c_1 e^{-x} + e^{\left(\frac{1}{2}\right)x} \left[ c_2 \cos\left(\frac{\sqrt{3}}{2}x\right) + c_3 \sin\left(\frac{\sqrt{3}}{2}x\right) \right]$$

5. Solve:  $(D^4 + 8D^2 + 16)y = 0$

**Solution:** The auxiliary equation is given by  $m^4 + 8m^2 + 16 = 0, (m^2 + 4)^2 = 0$

The roots are complex repeated  $m = \pm 2i, \pm 2i$ , hence the general solution is given by

$$y = [(c_1 + c_2 x) \cos(2x) + (c_3 + c_4 x) \sin(2x)]$$

6. Solve the initial value problem  $\frac{d^2x}{dt^2} + 5\frac{dx}{dt} + 6x = 0$ , given that  $x(0) = 0, \frac{dx}{dt}(0) = 15$ .

**Solution:** The auxiliary equation is given by  $m^2 + 5m + 6 = 0$

The roots are  $m = -3, -2$ , hence the general solution is given by  $x = c_1 e^{-3t} + c_2 e^{-2t}$

Using the given initial conditions, we find that

$$c_1 + c_2 = 0 \text{ and } 2c_1 + 3c_2 = -15$$

Solving these we get,  $c_1 = 15$  and  $c_2 = -15$ .

The particular solution of the given differential equation is  $x = 15(e^{-3t} - e^{-2t})$

### Exercise:

1 Solve  $(D^3 - 8)y = 0$ , where  $D \equiv \frac{d}{dx}$ .

$$\text{Answer: } y = c_1 e^{2x} + e^{-x} (c_2 \cos \sqrt{3}x + c_3 \sin \sqrt{3}x).$$

2 Solve  $\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y = 0; y(0) = 0, \frac{dy(0)}{dt} = 15$ .

$$\text{Answer: } y = 15(e^{-2t} - e^{-3t}).$$

3 Solve  $(D^4 - 4D^3 + 8D^2 - 8D + 4)y = 0$ , where  $D \equiv \frac{d}{dx}$ .

$$\text{Answer: } y = e^x [(c_1 + c_2 x) \cos x + (c_3 + c_4 x) \sin x].$$

4 Solve  $(D^3 - D^2 - D + 1)y = 0$ , where  $D \equiv \frac{d}{dx}$ .

$$\text{Answer: } y = (c_1 + c_2 x)e^x + c_3 e^{-x}.$$

5 Solve  $\frac{d^3y}{dx^3} - 2\frac{d^2y}{dx^2} + 4\frac{dy}{dx} - 8y = 0$ .

$$\text{Answer: } y = c_1 e^{2x} + c_2 \cos 2x + c_3 \sin 2x.$$

6 Find the solution of the boundary value problem  $y'' + y = 0$  given that  $y(0) =$

$$1, y\left(\frac{\pi}{2}\right) = 2.$$

$$\text{Answer: } y = \cos x + 2 \sin x.$$

- 7 Find the solution of the initial value problem:  $\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y = 0; y(0) = 0, \frac{dy(0)}{dt} = 15$ .

**Answer:**  $y = 15(e^{-2t} - e^{-3t})$ .

### Non-homogeneous Linear ODE with constant coefficients:

The general solution of non-homogeneous linear ODE has two parts –

Complementary function and Particular integral

The complementary function is solution of homogeneous ODE

The rules for finding particular integral are given below:

The kinds of functions that make up the input function  $g(x)$  are generally constants, exponentials, sine, cosine and polynomials.

$$\text{Consider } F(D)y = g(x),$$

If  $y = y_p$  is a particular integral of this ODE, then

$$F(D)y_p = g(x)$$

$$\text{then } y_p = \frac{1}{F(D)}g(x),$$

where  $\frac{1}{F(D)}$  is called as ‘inverse differential operator’.

### Rules for finding particular integral:

i) Let  $g(x) = ke^{ax}$

$$\text{then } y_p = k \frac{1}{F(D)}e^{ax} = k \frac{1}{F(a)}e^{ax}, \text{ provided } F(a) \neq 0$$

If  $F(a) = 0$ , then  $y_p = k \frac{x}{[F'(D)]_{D=a}}e^{ax}$ , provided  $F'(a) \neq 0$

If  $F'(a) = 0$ , then  $y_p = k \frac{x^2}{[F''(D)]_{D=a}}e^{ax}$ , provided  $F''(a) \neq 0$  and so on

ii) Let  $g(x) = \sin(ax + b)$  or  $\cos(ax + b)$

$$y_p = \frac{1}{F(D^2)}\sin(ax + b) \text{ or } \frac{1}{F(D^2)}\cos(ax + b)$$

$$= \frac{1}{F(-a^2)}\sin(ax + b) \text{ or } \frac{1}{F(-a^2)}\cos(ax + b), \text{ provided}$$

$$F(-a^2) \neq 0$$

If  $F(-a^2) = 0$ ,  $y_p = \frac{x}{F'(-a^2)}\sin(ax + b)$  or  $\frac{x}{F'(-a^2)}\cos(ax + b)$ , provided  $F'(-a^2) \neq 0$

Particular case:  $\frac{1}{(D^2+a^2)}\cos(ax + b) = \frac{x}{2a}\sin(ax + b)$

$$\frac{1}{(D^2+a^2)}\sin(ax + b) = -\frac{x}{2a}\cos(ax + b)$$

iii) Let  $g(x) = x^m$  then  $y_p = \frac{1}{F(D)}x^m = [F(D)]^{-1}x^m$

Expanding the right-hand side as a binomial series, the particular integral can be obtained. The following series expansions are useful:

$$(1 + D)^{-1} = 1 - D + D^2 - D^3 + \dots$$

$$(1 - D)^{-1} = 1 + D + D^2 + D^3 + \dots$$

$$(1 + D)^{-2} = 1 - 2D + 3D^2 - 4D^3 + \dots$$

$$(1 - D)^{-2} = 1 + 2D + 3D^2 + 4D^3 + \dots$$

iv) Let  $g(x) = e^{ax}V(x)$  where  $V(x)$  is function of ‘x’

$$y_p = \frac{1}{F(D)} e^{ax} V(x) = e^{ax} \frac{1}{F(D+a)} V(x)$$

The right-hand side can be evaluated by earlier methods.

Nonhomogeneous equations include external inputs (forcing functions), and the particular integral represents the system's response to these inputs. Concepts like inverse differential operators provide systematic ways to compute these responses. In computer science, these ideas are applied in signal processing, system design, and simulation, where systems respond to inputs such as signals, user actions, or external data streams, and their behaviour must be predicted or controlled.

### Examples:

1. Solve:  $(D - 2)^2 y = 8(e^{2x} + \sin 2x + x^2)$

**Solution:** The auxiliary equation is  $(m - 2)^2 = 0$ , the roots are  $m = 2, 2$

$$\therefore y_c = c_1 e^{2x} + c_2 x e^{2x}$$

The particular integral is obtained as follows:  $y_p = 8 \left[ \frac{1}{(D-2)^2} e^{2x} + \frac{1}{(D-2)^2} \sin 2x + \frac{1}{(D-2)^2} x^2 \right]$

$$\text{Now } \frac{1}{(D-2)^2} e^{2x} = x^2 \frac{1}{[2(1-0)]_{D=2}} e^{2x} = \frac{x^2 e^{2x}}{2}$$

$$\begin{aligned} \frac{1}{(D-2)^2} \sin 2x &= \frac{1}{(D^2-D+4)} \sin 2x = \frac{1}{-4-4D+4} \sin 2x \\ &= -\frac{1}{4D} \sin 2x = -\frac{1}{4} \int \sin 2x \, dx = \frac{1}{8} \cos 2x \end{aligned}$$

$$\frac{1}{(D-2)^2} x^2 = \frac{1}{4} \left( -\frac{D}{2} \right)^{-2} x^2 = \frac{1}{4} \left( 1 + \frac{2D}{2} + \frac{3D^2}{4} + \dots \right) x^2 = \frac{1}{4} \left( x^2 + 2x + \frac{3}{2} \right)$$

$$\therefore y_p = 4x^2 e^{2x} + \cos 2x + 2x^2 + 4x + 3$$

Hence, the general solution of given equation is  $y = y_c + y_p$

$$y = c_1 e^{2x} + c_2 x e^{2x} + 4x^2 e^{2x} + \cos 2x + 2x^2 + 4x + 3$$

2. Solve:  $y'' - 4y = \cosh(2x - 1) + 3^x$

**Solution:** The auxiliary equation and its roots are:  $m^2 - 4 = 0$   $m = +2, -2$

$$\therefore y_c = c_1 e^{-2x} + c_2 e^{2x}$$

The particular integral is obtained as follows:

$$y_p = \frac{1}{(D^2 - 4)} \cosh(2x - 1) + \frac{1}{(D^2 - 4)} 3^x$$

$$\text{Now } \frac{1}{(D^2 - 4)} \cosh(2x - 1) = \frac{1}{2} \left[ \frac{e^{2x-1} + e^{-2x+1}}{D^2 - 4} \right] = \frac{1}{8} (x e^{2x-1} - x e^{-2x+1})$$

$$\frac{1}{(D^2 - 4)} 3^x = \frac{1}{(D^2 - 4)} e^{x \log(3)} = \frac{1}{[(\log 3)^2 - 4]} 3^x$$

Hence, the solution of given equation is  $y = y_c + y_p$

$$\therefore y = c_1 e^{-2x} + c_2 e^{2x} + \frac{1}{8} (x e^{2x-1} - x e^{-2x+1}) + \frac{1}{[(\log 3)^2 - 4]} 3^x$$

3. Solve:  $y'' - 2y' + y = x e^x \sin x$

**Solution:** The auxiliary equation and its roots are:  $m^2 - 2m + 1 = 0$   $m = 1, 1$

$$\therefore y_c = c_1 e^x + c_2 x e^x$$

The particular integral is obtained as follows:

$$y_p = \left[ \frac{1}{(D-1)^2} \right] e^x x \sin x = e^x \frac{1}{[D+1-1]^2} x \sin x = e^x \frac{1}{D^2} x \sin x$$

Integrating right hand side twice by using the rule of parts we obtain:

$$y_p = -e^x (x \sin x + 2 \cos x)$$

Hence, the solution of given equation is  $y = y_c + y_p$

$$\therefore y = c_1 e^x + c_2 x e^x - e^x (x \sin x + 2 \cos x)$$

4. Solve:  $(D^3 + 3D^2 + 3D + 1)y = e^{-x}$

**Solution:** The auxiliary equation and its roots are:

$$m^3 + 3m^2 + 3m + 1 = 0$$

$$m = -1, -1, -1$$

$$\therefore y_c = c_1 e^{-x} + c_2 x e^{-x} + c_3 x^2 e^{-x}$$

$$\text{Now } \frac{1}{(D+1)^3} e^{-x} = x^2 \frac{1}{[6(D+1)]_{D=-1}} e^{-x} = \frac{x^3 e^{-x}}{6}$$

Hence, the solution of given equation is  $y = y_c + y_p$

$$\therefore y = c_1 e^{-x} + c_2 x e^{-x} + c_3 x^2 e^{-x} + \frac{x^3 e^{-x}}{6}$$

5. Find the particular integral of  $(D^3 + 1)y = \cos(2x - 1)$ .

**Solution:** The particular integral is given by:

$$y_p = \left[ \frac{1}{D^3+1} \right] \cos(2x-1) = \frac{1}{D(-4)+1} \cos(2x-1) \\ = \frac{(1+4D)}{1-16D^2} \cos(2x-1)$$

$$y_p = \frac{(1+4D)}{1-16(-4)} \cos(2x-1) = \frac{1}{65} [\cos(2x-1) - 8 \sin(2x-1)]$$

6. Find the particular integral of  $(D^3 + 4D)y = 2 \sin x \cos x$ .

**Solution:** The particular integral is given by:

$$y_p = \left[ \frac{1}{D^3 + 4D} \right] 2 \sin x \cos x = \frac{1}{(D^2 + 4)D} \sin(2x)$$

$$y_p = x \frac{1}{3D^2 + 4} \sin(2x) = x \frac{1}{3(-4) + 4} \sin(2x) = -\frac{x}{8} \sin(2x)$$

7. Find the particular integral of  $(D^2 + 9)y = x \cos x$

**Solution:** The particular integral is given by:

$$y_p = \frac{1}{D^2 + 9} \operatorname{Re}(xe^{ix})$$

$$= \operatorname{Re} \frac{1}{D^2 + 9} (xe^{ix})$$

$$= \operatorname{Re} e^{ix} \frac{1}{(D + i)^2 + 9} (x)$$

$$y_p = \operatorname{Re} e^{ix} \frac{1}{D^2 + 2iD + 8} (x)$$

$$= \operatorname{Re} e^{ix} \frac{1}{8 \left(1 + \frac{iD}{4}\right)} (x)$$

$$= \operatorname{Re} \frac{e^{ix}}{8} \left(1 + \frac{iD}{4}\right)^{-1} (x)$$

$$y_p = \operatorname{Re} \frac{e^{ix}}{8} \left(1 - \frac{iD}{4}\right) (x)$$

$$= \operatorname{Re} \frac{\cos x + i \sin x}{8} \left(x - \frac{i}{4}\right)$$

$$= \frac{1}{32} (4x \cos x + \sin x)$$

8. Find the particular integral of  $(D^2 - 4D + 4)y = x^2 e^{3x} + \sin^2 x$

**Solution:** The particular integral is given by:

$$y_p = \frac{1}{(D-2)^2} [x^2 e^{3x} + \sin^2 x]$$

$$= \frac{1}{(D-2)^2} [x^2 e^{3x}] + \frac{1}{(D-2)^2} [\sin^2 x]$$

Now,  $\frac{1}{(D-2)^2} [x^2 e^{3x}]$

$$= e^{3x} \frac{1}{(D + 3 - 2)^2} [x^2]$$

$$= e^{3x} \frac{1}{(D + 1)^2} [x^2]$$

$$= e^{3x} (1 + D)^{-2} [x^2]$$

$$= e^{3x} (1 - 2D + 3D^2) [x^2]$$

$$= e^{3x} (x^2 - 4x + 6)$$

and

$$\frac{1}{(D-2)^2} [\sin^2 x]$$



$$\begin{aligned}
 &= \frac{1}{(D-2)^2} \left[ \frac{1 - \cos 2x}{2} \right] \\
 &= \frac{1}{(D-2)^2} \left[ \frac{1}{2} e^{0x} \right] - \frac{1}{(D-2)^2} \left[ \frac{1}{2} \cos 2x \right] \\
 &= \frac{1}{(0-2)^2} \left[ \frac{1}{2} e^{0x} \right] - \frac{1}{D^2 - 4D + 4} \left[ \frac{1}{2} \cos 2x \right] \\
 &= \frac{1}{8} - \frac{1}{2} \cdot \frac{1}{2^2 - 4D + 4} [\cos 2x] \\
 &= \frac{1}{8} + \frac{1}{8D} [\cos 2x] = \frac{1}{8} + \frac{1}{16} \sin 2x \\
 \therefore y_p &= e^{3x}(x^2 - 4x + 6) + \frac{1}{8} + \frac{1}{16} \sin 2x
 \end{aligned}$$

**Exercise:**

1. Solve  $\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + y = 2e^{2t}$ .

**Answer:**  $y = (c_1 + c_2 t)e^{-t} + \frac{2}{9}e^{2t}$

2. Solve  $y''' - 12y' + 16y = (e^x + e^{-2x})^2$ .

**Answer:**  $y = (c_1 + c_2 x)e^{2x} + c_3 e^{-4x} + \frac{x^2}{12}e^{2x} + \frac{2}{27}e^{-x} + \frac{x}{36}e^{-4x}$

3. Solve  $\frac{d^3y}{dx^3} + \frac{d^2y}{dx^2} - \frac{dy}{dx} - y = \cos 2x$ .

**Answer:**  $y = ae^x + (b + cx)e^{-x} - \frac{1}{25}(2 \sin 2x + \cos 2x)$

4. Solve  $(D^3 - D^2 - 6D)x = 1 + t^2$ .

**Answer:**  $x = a + be^{3t} + ce^{-2t} - \frac{1}{108}(25t - 3t^2 + 6t^3)$

5. Solve  $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = e^{2x} \sin x$ .

**Answer:**  $y = ae^{-x} + be^{-2x} - \frac{1}{170}e^{2x}(7 \cos x - 11 \sin x)$

6. Solve  $(D^4 + 4)y = \sin^2 x$ .

**Answer:**  $y = a \cos 2x + b \sin 2x + \frac{1}{8}(1 - x \sin 2x)$

7. Solve  $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = e^{3x}(x^2 + 7x + 5)$ .

**Answer:**  $y = \left( a + bx + \frac{1}{12}(x^4 + 14x^3 + 30x^2) \right) e^{3x}$

8. Solve  $\frac{d^2y}{dx^2} - 4y = x \sinh x$ .

**Answer:**  $y = ae^{2x} + be^{-2x} - \frac{x}{3} \sinh x - \frac{2}{9} \cosh x$

9. Solve  $(D^2 - 4D + 4)y = 8x^2 e^{2x} \sin 2x$ .

**Answer:**  $y = (a + bx(3 - 2x^2) \sin 2x - 4x \cos 2x) e^{2x}$

10. Solve  $y'' - y = x \sin x + (1 + x^2)e^x$ .

**Answer:**  $y = c_1 e^{-x} + c_2 e^x - \frac{x}{2} \sin x - \frac{1}{2} \cos x + \frac{e^x}{12} \left( \frac{3x}{2} - \frac{x^2}{2} + \frac{x^3}{3} \right)$

### Method of Variation of Parameters:

The method of variation of parameter is more general method, in which the function  $g(x)$  is not restricted to any particular form. The working rule for finding solution of the given ODE is given below:

Consider the second order ODE of the form  $y'' + P(x)y' + Q(x)y = g(x)$

Let  $y = c_1y_1 + c_2y_2$  be solution of the equation with  $g(x) = 0$ , the complete solution of given ODE is  $y = A(x)y_1 + B(x)y_2$ , where

$$A(x) = - \int \frac{y_2 g(x)}{W} dx + k_1$$

$$B(x) = \int \frac{y_1 g(x)}{W} dx + k_2$$

With the 'Wronskian' given by  $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} \neq 0$

Method of variation of parameters provides a general technique to solve nonhomogeneous differential equations when simpler methods fail. In computer science, it is relevant in advanced simulations, control theory, and numerical analysis, where flexible solution techniques are required for complex dynamic models, especially when dealing with non-standard or time-varying inputs.

### Examples:

1. **Solve:**  $y'' + a^2y = \sec(ax)$

**Solution:** The auxiliary equation and its roots are  $m^2 + a^2 = 0$ ,  $m = \pm ai$

$$\therefore y_c = c_1 \cos(ax) + c_2 \sin(ax)$$

$$\text{Let } y_1 = \cos(ax), y_2 = \sin(ax)$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = a[\cos^2(ax) + \sin^2(ax)] = a$$

$$A(x) = - \int \frac{y_2 g(x)}{W} dx + k_1 = - \int \frac{\tan(ax)}{a} dx + k_1 = - \frac{1}{a^2} \log[\cos(ax)] + k_1$$

$$B(x) = \int \frac{y_1 g(x)}{W} dx + k_2 = \frac{1}{a} \int dx + k_2 = \frac{x}{a} + k_2$$

The complete solution of given ODE is  $y = A(x)y_1 + B(x)y_2$ ,

$$\therefore y = k_1 \cos(ax) + k_2 \sin(ax) - \frac{1}{a^2} \cos(ax) \log[\cos(ax)] + \frac{x}{a} \sin(ax)$$

2. **Solve:**  $y'' - y = e^{-2x} \cos(e^{-x})$

**Solution:** The auxiliary equation and its roots are  $m^2 - 1 = 0$ ,  $m = \pm 1$

$$\therefore y_c = c_1 e^{-x} + c_2 e^x$$

$$\text{Let } y_1 = e^x, y_2 = e^{-x}$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = -2$$

$$\begin{aligned} A(x) &= - \int \frac{y_2 g(x)}{W} dx + k_1 \\ &= - \frac{1}{2} \int z^2 \cos(z) dz + k_1, \end{aligned}$$

Where  $z = e^{-x}$ , and  $dz = -e^{-x} dx$

Integrating by rule of parts obtain:

$$\begin{aligned} A(x) &= -\frac{1}{2} e^{-2x} \sin(e^{-x}) - e^{-x} \cos(e^{-x}) + \sin(e^{-x}) + k_1 \\ B(x) &= \int \frac{y_1 g(x)}{W} dx + k_2 \\ &= \frac{1}{2} \int \cos(z) dz + k_2 = \frac{1}{2} \sin(e^{-x}) + k_2, \text{ where } z = e^{-x} \end{aligned}$$

The complete solution of given ODE is  $y = A(x)y_1 + B(x)y_2$

$$\therefore y = k_1 e^{-x} + k_2 e^x - \cos(e^{-x}) + e^x \sin(e^{-x})$$

### Exercise:

- Using the method of variation of parameters, solve  $(D^2 + 4)x = \tan 2t$ .

**Answer:**  $x = a \cos 2t + b \sin 2t - \frac{1}{4} \cos 2t \log(\sec 2t + \tan 2t)$

- Solve  $\frac{d^2 y}{dx^2} - y = \frac{2}{1+e^x}$  by the method of variation of parameters.

**Answer:**  $y = ae^x + be^{-x} - 1 + e^x \log(e^{-x} + 1) - e^{-x} \log(e^x + 1)$ .

- Solve  $y'' - 2y' + y = e^x \log x$  by the method of variation of parameters.

**Answer:**  $y = (a + bx)e^x + \frac{1}{4} x^2 e^x (2 \log x - 3)$ .

### Cauchy-Euler equation:

The linear ODE of the form

$$(a_0 x^n D^n + a_1 x^{n-1} D^{n-1} + a_2 x^{n-2} D^{n-2} + \dots + a_{n-1} x D + a_n) y = g(x)$$

where  $a_0, a_1, \dots, a_n$  are constants, is known as 'Cauchy-Euler' or equidimensional equation.

This equation can be reduced to ODE with constant coefficients by changing the independent variable as follows:

$$\begin{aligned} \text{Take } x &= e^z, \text{ then } xDy = D_1 y, \\ x^2 D^2 y &= D_1(D_1 - 1)y, \\ x^3 D^3 y &= D_1(D_1 - 1)(D_1 - 2)y \\ \text{where } D_1 &= \frac{d}{dz} \text{ and so on.} \end{aligned}$$

The resulting ODE can be solved using the earlier methods.

Cauchy-Euler equations involve variable coefficients and arise in systems where behavior scales with time or space. In computer science, they appear in algorithm analysis, scaling models, and computational physics, where problems involve non-uniform growth or scaling,

such as recursive algorithms, fractal modelling, and simulations with non-constant parameters.

### Examples:

1. Solve:  $4x^2 \frac{d^2y}{dx^2} + 8x \frac{dy}{dx} + y = 0$

**Solution:** The given ODE is second order Euler equation.

Put  $x = e^z$ , then  $x Dy = D_1 y$ ,  $x^2 D^2 y = D_1(D_1 - 1)y$

The given equation reduces to  $(D_1^2 + 4D_1 + 1)y = 0$

The auxiliary equation and its roots are  $m^2 + 4m + 1 = 0$ ,

$$m = -\frac{1}{2}, -\frac{1}{2}$$

The general solution of given ODE is  $y = c_1 e^{-(\frac{1}{2})z} + c_2 z e^{-(\frac{1}{2})z}$ , where  $z = \log(x)$

2. Solve:  $x^2 \frac{d^2y}{dx^2} - x \frac{dy}{dx} + y = \log(x)$

**Solution:** The given ODE is second order Euler equation.

Put  $x = e^z$ , then  $x Dy = D_1 y$ ,  $x^2 D^2 y = D_1(D_1 - 1)y$

The given equation reduces to  $(D_1^2 - 2D_1 + 1)y = z$

The auxiliary equation and its roots are  $m^2 - 2m + 1 = 0$ ,  $m = 1, 1$

$$\therefore y_c = c_1 e^z + c_2 z e^z$$

Now  $y_p = \frac{1}{(D_1 - 1)^2} z = (1 - D_1)^{-2} z = (1 + 2D_1 + 3D_1^2 + \dots)z = z + 2$

Hence, the solution of given equation is  $y = y_c + y_p$

$$y = c_1 e^z + c_2 z e^z + z + 2, \text{ where } z = \log(x)$$

### Exercise:

1. Solve  $4x \frac{d^2y}{dt^2} + 4 \frac{dy}{dt} - \frac{y}{x} = 4x$ .

**Answer:**  $y = a\sqrt{x} + b \frac{1}{\sqrt{x}} + \frac{4}{15} x^2$

2. Solve  $x^3 \frac{d^3y}{dx^3} + 2x \frac{dy}{dx} - 2y = x^2 \log x + 3x$

**Answer:**  $y = ax + x(b \cos \log x + c \sin \log x) - \frac{1}{2} x^2 + 3x \log x$

3. Solve  $(x^2 D^2 + xD + 9)y = 3x^2 + \sin(3 \log x)$ , where  $D \equiv \frac{d}{dx}$ .

**Answer:**  $y = a \cos(3 \log x) + b \sin(3 \log x) + \frac{3}{13} x^2 - \frac{1}{6} \log x \cos(3 \log x)$ .

### Applications of differential equations:

Simple harmonic motion models oscillatory systems, while LRC circuits describe electrical systems with resistance, inductance, and capacitance. In computer science, these are directly applied in signal processing, electronics simulation, and embedded systems, where oscillations, filtering, and resonance behaviour are important. They are also used in audio processing, communication systems, and hardware modelling, forming a bridge between mathematical models and real-world computational applications.

A flexible spring is suspended vertically from a rigid support and then a mass is attached to its free end. Assuming that the mass vibrates, free of other external forces the displacement describes SHM. Fig (1)

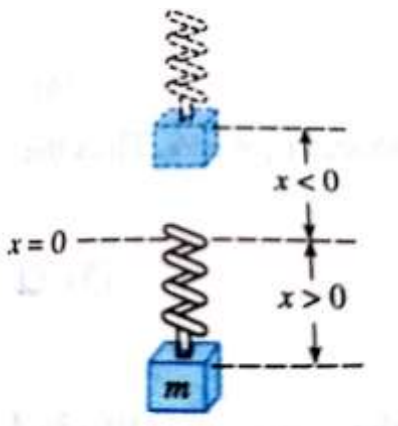


Fig (1)

1. The displacement 'x' of a particle executing simple harmonic motion (SHM) is given by  $\ddot{x} = -n^2x$ . Where  $n^2$ , is a force constant. Solve this equation under the initial conditions  $x = a$  and  $\frac{dx}{dt} = 0$ , at  $t = 0$ .

**Solution:** The given ODE is  $(D_1^2 + n^2)x = 0$ , where  $D_1 = \frac{d}{dt}$

The auxiliary equation and its roots are  $m^2 - n^2 = 0$

$$m = \pm ni$$

The general solution is given by  $x = c_1 \cos(nt) + c_2 \sin(nt)$ , using the given conditions  $x = a$  and  $\frac{dx}{dt} = 0$ , at  $t = 0$  we find that  $c_1 = a$  and  $c_2 = 0$

The displacement at any instant of time is given by  $x = a \cos(nt)$

2. The charge  $q = q(t)$  on the capacitor in a LRC series circuit (Fig (2)) is governed by the equation  $L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = E(t)$ . Find the charge on the capacitor  $q = q(t)$ ,

when  $L=0.25H$ ,  $R=10\Omega$ ,  $C=0.001F$ ,  $E(t) = 0$ ,  $q(0) = q_0$ ,

$\frac{dq}{dt} = 0$ , for  $t = 0$ .

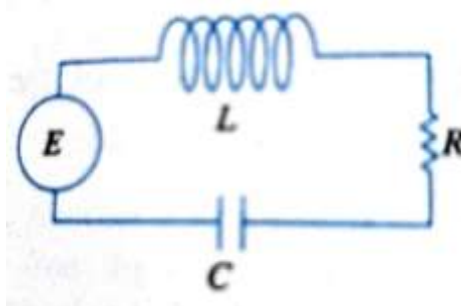


Fig (2)

**Solution:** The given ODE is  $\frac{1}{4} \frac{d^2q}{dt^2} + 10 \frac{dq}{dt} + 1000q = 0$

$$\text{or } \frac{d^2q}{dt^2} + 40 \frac{dq}{dt} + 4000q = 0$$

The auxiliary equation and its roots are  $m^2 + 40m + 4000 = 0$ ,  
 $m = -20 \pm 60i$

The general solution is given by  $q = q(t)$   
 $= e^{-20t} [c_1 \cos(60t) + c_2 \sin(60t)]$

Using the given conditions, we get  $c_1 = q_0$  and  $c_2 = q_0/3$

The charge on the capacitor is given by:

$$q = q(t) = e^{-20t} [q_0 \cos(60t) + \frac{q_0}{3} \sin(60t)]$$

3. Find the charge on the capacitor  $q = q(t)$ , in a LRC series circuit from

$$L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = E(t),$$

when  $L=(5/3)H$ ,  $R=10\Omega$ ,  $C=(1/30)F$ ,  $E(t) = 300V$ ,  $q(0) = 0$ ,  $\frac{dq}{dt} = 0$

**Solution:** The given ODE is  $\frac{5}{3} \frac{d^2q}{dt^2} + 10 \frac{dq}{dt} + 30q = 300$  or

$$5 \frac{d^2q}{dt^2} + 30 \frac{dq}{dt} + 90q = 900$$

The auxiliary equation and its roots are  $5m^2 + 30m + 90 = 0$ ,  
 $m = -3 \pm 3i$

$$q_p = \left[ \frac{1}{5D_1^2 + 30D_1 + 90} \right] 900e^{0t} = \frac{900}{90} = 10$$

The general solution is given by:

$$q = q(t) = e^{-3t} [c_1 \cos(3t) + c_2 \sin(3t)] + 10$$

Using the given conditions, we get  $c_1 = -10$  and  $c_2 = -10$

The charge on the capacitor is given by:

$$q = q(t) = 10 - e^{-20t} [\cos(3t) + \sin 3t]$$

### Video links:

[https://www.youtube.com/watch?v=V9bl02Ffo\\_o](https://www.youtube.com/watch?v=V9bl02Ffo_o)